

Math 526, F06
Quiz 7

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (12 points) Consider the system $\mathbf{Ax} = \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & -4 & 2 \\ 2 & -1 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ -1 \\ 5 \end{bmatrix}$$

a) Write down the corresponding $\mathbf{S}$, $\mathbf{T}$ and the iterative formula for the Jacobi method.

Solution: The Jacobi method:

$$\mathbf{Sx}^{new} = \mathbf{b} + \mathbf{Tx}^{old}$$
$$\mathbf{S} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \quad \mathbf{T} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & -2 \\ -2 & 1 & 0 \end{bmatrix}$$

i.e.,

$$\begin{aligned} x_1^{new} &= (-1 - x_2^{old})/2 \\ x_2^{new} &= (-1 - x_1^{old} - 2x_3^{old})/(-4) \\ x_3^{new} &= (5 - 2x_1^{old} + x_2^{old})/4 \end{aligned}$$

b) Starting with the initial guess $\mathbf{x}_0 = \mathbf{0}$, compute $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ using the Jacobi method. Also compute $\|\mathbf{x} - \mathbf{x}_3\|_\infty$ where $\mathbf{x} = [-1, 1, 2]^t$ is the exact solution.

Solution: Using $\mathbf{Sx}_{k+1} = \mathbf{b} + \mathbf{Tx}_k$ from a), we get

$$\begin{aligned} \mathbf{x}_1 &= \begin{bmatrix} (-1 - 0)/2 \\ (-1 - 0 - 0)/(-4) \\ (5 - 0 + 0)/4 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/4 \\ 5/4 \end{bmatrix} \\ \mathbf{x}_2 &= \begin{bmatrix} (-1 - 1/4)/2 \\ (-1 + 1/2 - 5/2)/(-4) \\ (5 + 1 + 1/4)/4 \end{bmatrix} = \begin{bmatrix} -5/8 \\ 3/4 \\ 25/16 \end{bmatrix} \\ \mathbf{x}_3 &= \begin{bmatrix} (-1 - 3/4)/2 \\ (-1 + 5/8 - 25/8)/(-4) \\ (5 + 5/4 + 3/4)/4 \end{bmatrix} = \begin{bmatrix} -7/8 \\ 7/8 \\ 7/4 \end{bmatrix} \end{aligned}$$

Then it is easy to find that

$$\|\mathbf{x} - \mathbf{x}_3\|_\infty = 1/4$$

Problem 2. (8 points) Let $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$. Find $\kappa_1(\mathbf{A})$ and $\kappa_\infty(\mathbf{A})$.

Solution: Using the formula in Exercise 6.4, it is easy to find that

$$\mathbf{A}^{-1} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}$$

Then, we have

$$\|\mathbf{A}\|_1 = \max\{2+3, 3+5\} = 8, \quad \|\mathbf{A}^{-1}\|_1 = \{5+|-3|, |-3|+2\} = 8,$$

$$\|\mathbf{A}\|_\infty = \max\{2+3, 3+5\} = 8, \quad \|\mathbf{A}^{-1}\|_\infty = \{5+|-3|, |-3|+2\} = 8.$$

So

$$\kappa_1(A) = \|\mathbf{A}\|_1 \|\mathbf{A}^{-1}\|_1 = 8 \times 8 = 64$$

$$\kappa_\infty(A) = \|\mathbf{A}\|_\infty \|\mathbf{A}^{-1}\|_\infty = 8 \times 8 = 64$$